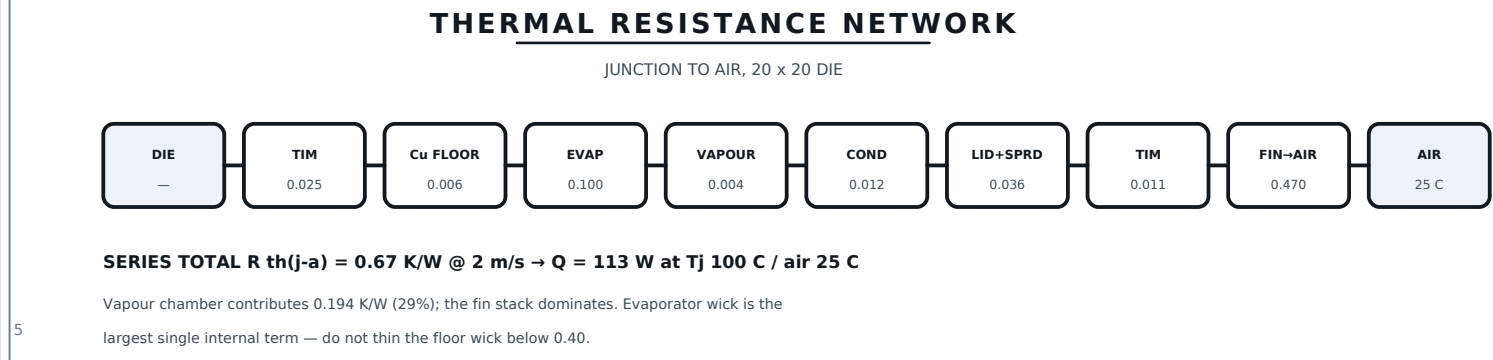
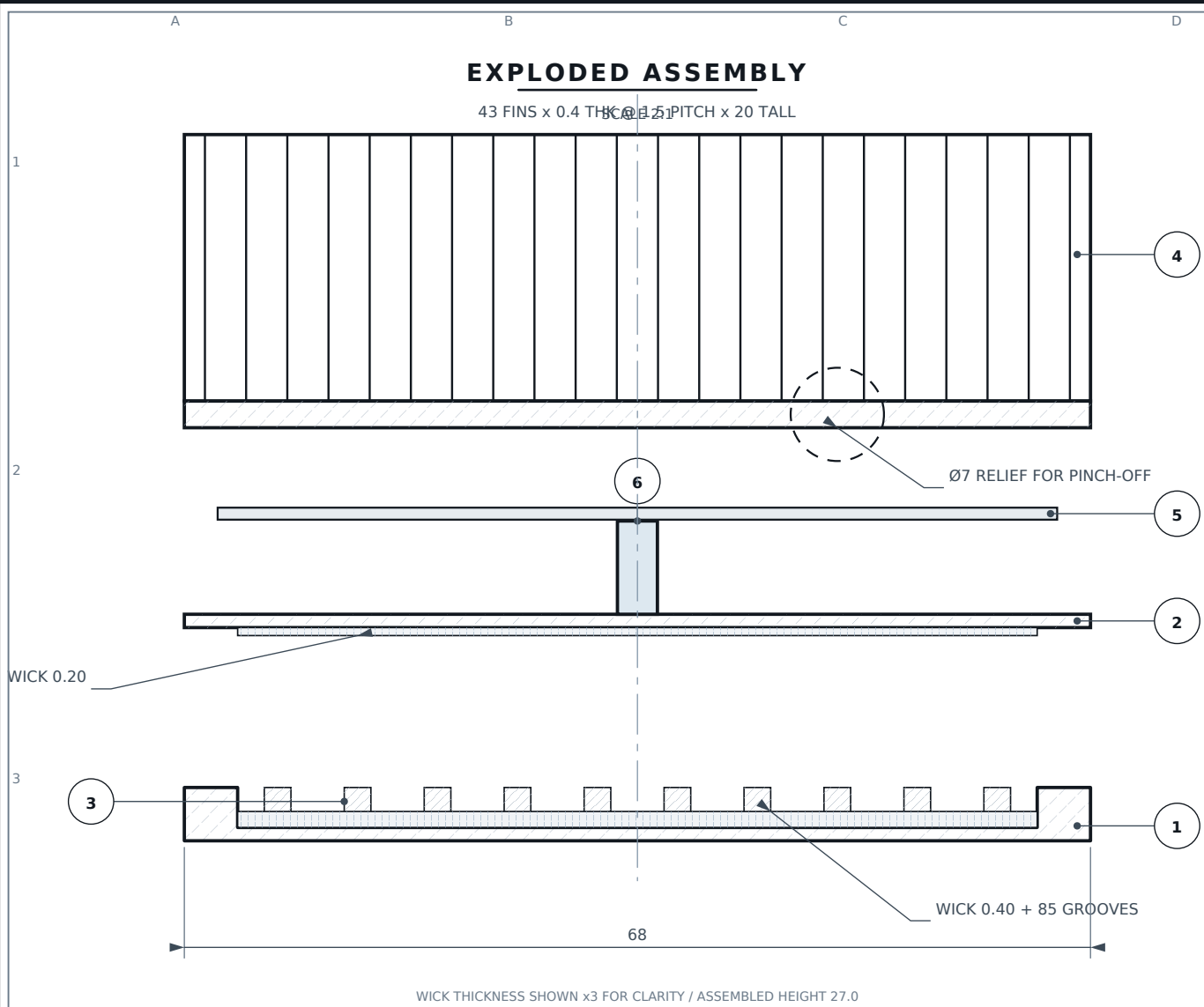


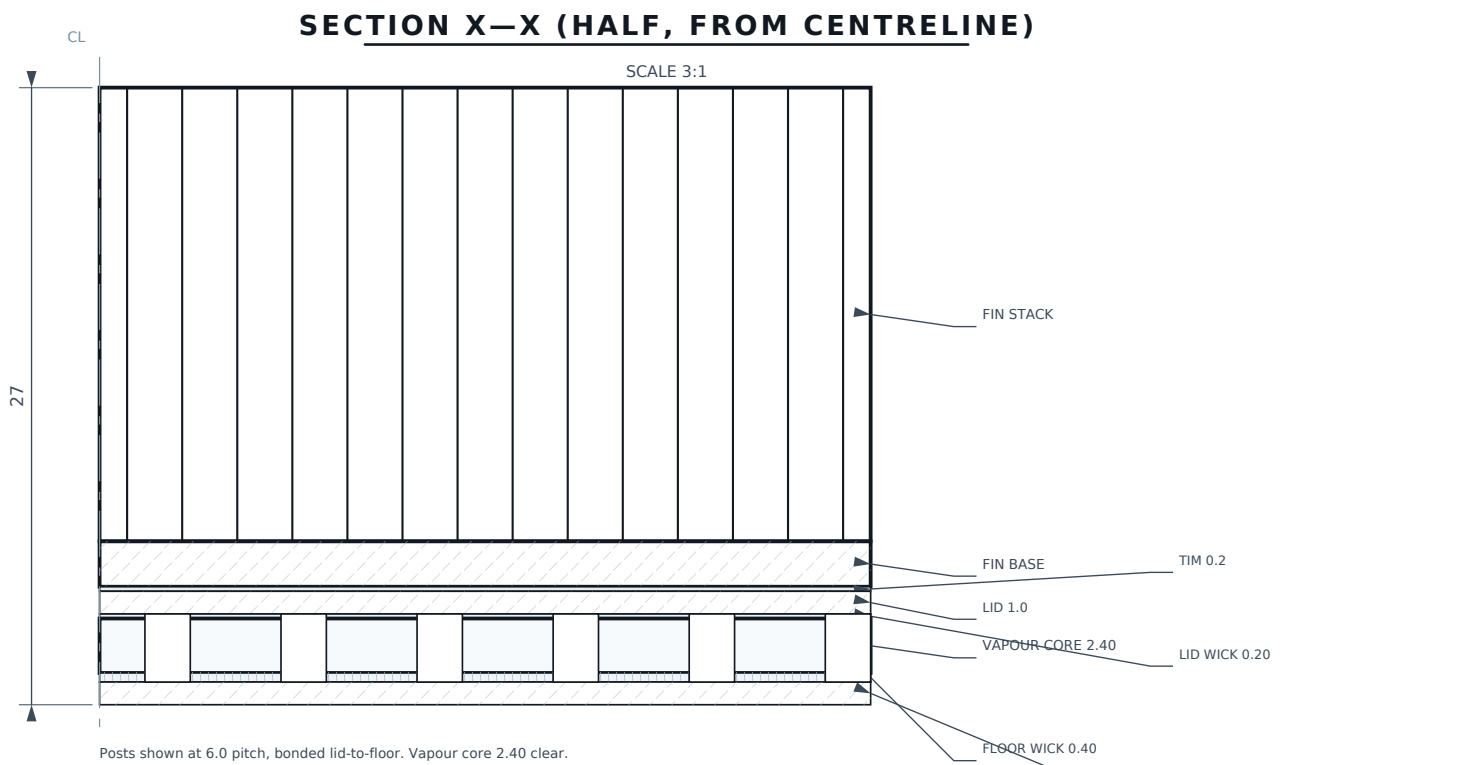


- GENERAL NOTES**
- Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
  - Charging, evacuation and sealing per VC-300. NEVER charge before diffusion bonding — bond temperature exceeds the fluid critical pressure envelope.
  - CONFIG A requires forced air  $\geq 1 \text{ m/s}$ . For still air use CONFIG B (VC-202); a 1.5 mm fin pitch chokes natural convection to  $h = 0.03 \text{ W/m}^2\text{K}$ .
  - Orientation-independent: 10.5 kPa capillary head vs 0.66 kPa gravity head over 68 mm.

| BILL OF MATERIALS — ASSEMBLY VC-100 |         |          |                                     |                   |        |
|-------------------------------------|---------|----------|-------------------------------------|-------------------|--------|
| ITEM                                | QTY     | PART No. | DESCRIPTION                         | MATERIAL          | MASS   |
| 1                                   | 1       | VC-101   | CHAMBER BASE, POCKETED, 100 POSTS   | Cu C1100 OFHC     | 77.4 g |
| 2                                   | 1       | VC-102   | CHAMBER LID, 1.0 THK, FILL PORT     | Cu C1100 OFHC     | 41.4 g |
| 3                                   | 1       | VC-103   | SINTERED WICK — IN-SITU PROCESS     | Cu POWDER 45-75um | 9.6 g  |
| 4                                   | 1       | VC-201   | FIN STACK, 43 FIN — FORCED AIR      | Al 1100-H14       | 88.1 g |
| 4A                                  | 1       | VC-202   | FIN STACK, 13 FIN — NAT. CONVECTION | Al 1100-H14       | 72.7 g |
| 5                                   | 1       | VC-204   | THERMAL INTERFACE PAD 0.2 THK       | PCM / SIL-PAD     | 0.4 g  |
| 6                                   | 1       | VC-205   | PINCH-OFF TUBE OD3.0 x ID2.0 x 25   | Cu C1100          | 1.2 g  |
| 7                                   | 1.90 mL | VC-206   | WORKING FLUID — DEAERATED           | DI H2O 18 MOhm.cm | 1.87 g |

TOTAL MASS, CHARGED: 218 g (CONFIG A) / 203 g (CONFIG B)

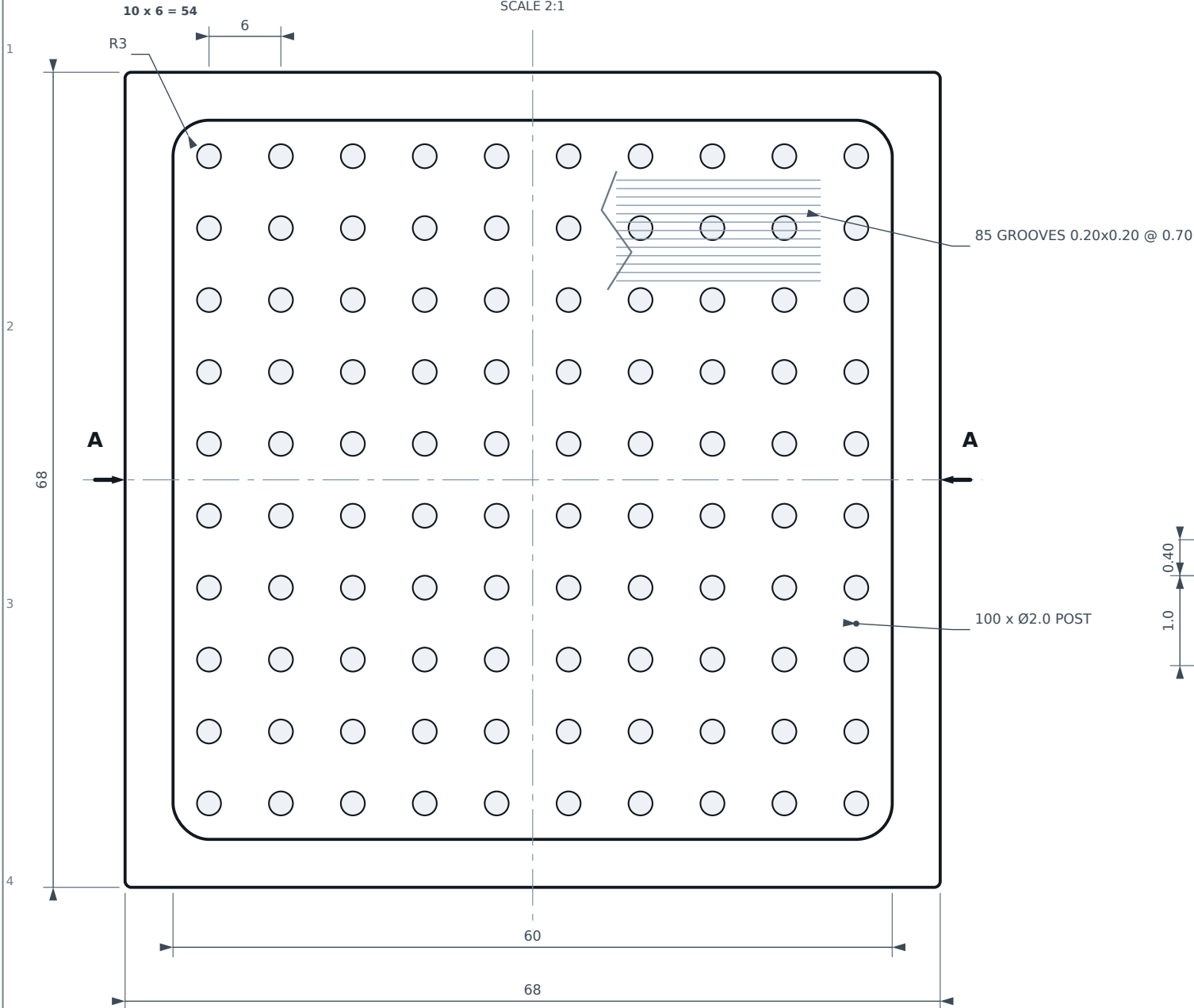
| PERFORMANCE — 20 x 20 DIE, $T_j 100 \text{ C}$ , AIR IN 25 C |         |               |                       |
|--------------------------------------------------------------|---------|---------------|-----------------------|
| CONFIGURATION                                                | AIR     | $R_{th(j-a)}$ | $Q @ dT 75 \text{ K}$ |
| CONFIG A — 43 fin @ 1.5 pitch                                | 1 m/s   | 1.12 K/W      | 67 W                  |
| CONFIG A — 43 fin @ 1.5 pitch                                | 2 m/s   | 0.67 K/W      | 113 W                 |
| CONFIG A — 43 fin @ 1.5 pitch                                | 4 m/s   | 0.46 K/W      | 163 W                 |
| CONFIG B — 13 fin @ 5.0 pitch                                | natural | 3.42 K/W      | 22 W                  |
| CONFIG B — passive, $T_j 85 \text{ C limit}$                 | natural | —             | 16.5 W                |
| capillary limit — water, vertical                            | —       | —             | 804 W                 |
| capillary limit — water, horizontal                          | —       | —             | 858 W                 |



| INDIA-VC THERMAL HARDWARE                                                              |             |          |                        |  | ISO 128 / ISO 2768-mK |  |
|----------------------------------------------------------------------------------------|-------------|----------|------------------------|--|-----------------------|--|
| TITLE<br><b>GENERAL ASSEMBLY</b><br>VAPOUR CHAMBER + FIN STACK / Cu-H2O / 68 x 68 x 27 |             |          | SCALE<br><b>2:1</b>    |  | UNITS<br><b>mm</b>    |  |
|                                                                                        |             |          | THIRD ANGLE PROJECTION |  |                       |  |
| DRAWING No. <b>VC-100</b>                                                              |             |          | REV <b>A</b>           |  | SHEET <b>1 / 7</b>    |  |
| DRAWN                                                                                  | CHECKED     | MATERIAL | FINISH                 |  | DATE                  |  |
| AI-ASSISTED DRAFT                                                                      | — PENDING — | SEE BOM  | SEE NOTES              |  | 2026-09-09            |  |

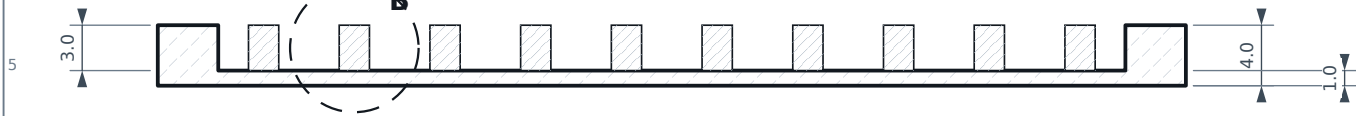
PLAN — POCKET SIDE UP

SCALE 2:1



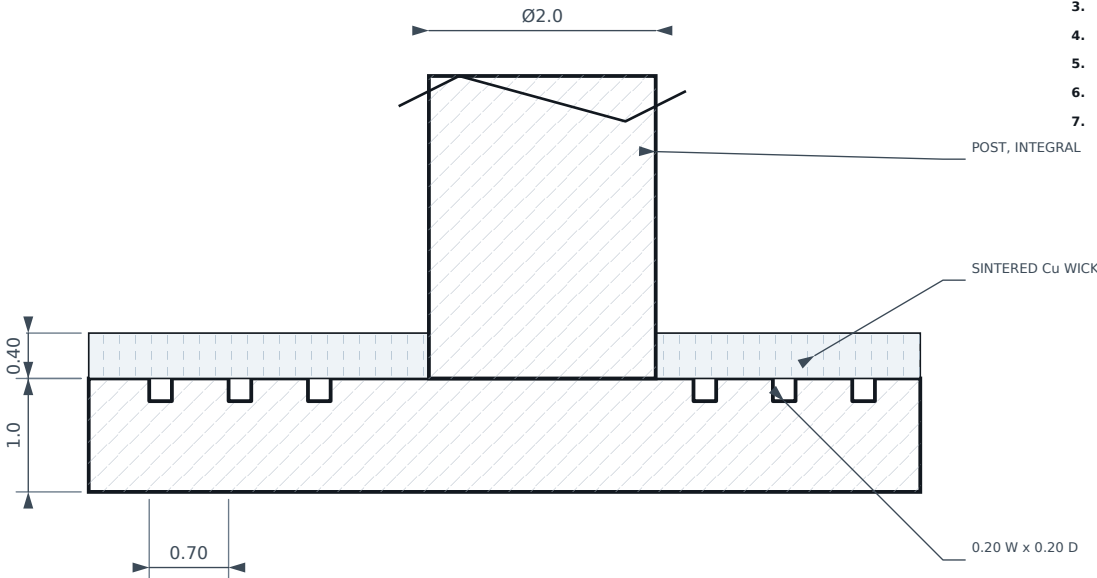
SECTION A—A

SCALE 2:1



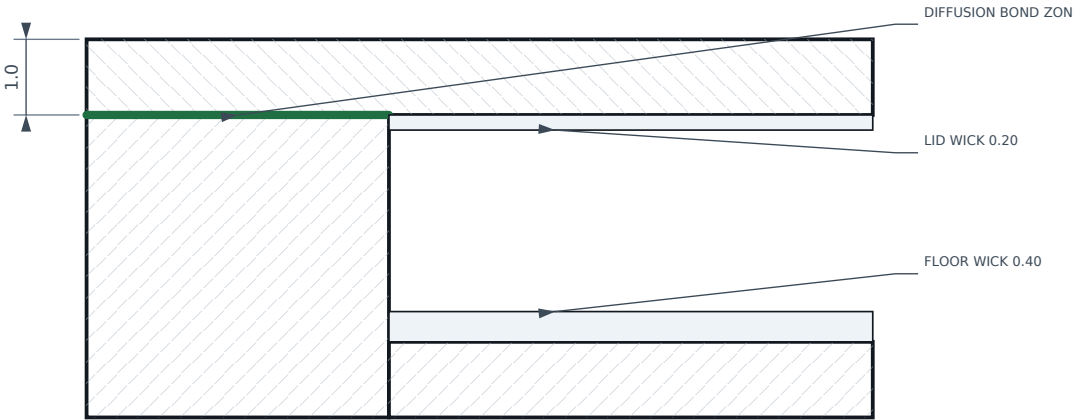
DETAIL B

SCALE 15:1



DETAIL C — RIM BOND LAND

Bond area 4.0 x 4 sides = 16.0 mm<sup>2</sup>. Burst margin > 16 at 150 C.



NOTES

- MATL: Cu C1100 / C10100 OFHC, annealed.
- Posts integral with base — machine from solid or coin.
- Bond faces (rim + post tops): flatness 0.02, Ra 0.4 max.
- Grooves run ONE direction only, interrupted at posts.
- Post top faces coplanar with rim within 0.015.
- Deburr 0.1 max. No burrs into cavity.
- Vacuum degrease + acid pickle before sintering.

CRITICAL CHARACTERISTICS

| FEATURE                    | NOMINAL       | TOL           | CRITICAL TO        |
|----------------------------|---------------|---------------|--------------------|
| Floor thickness            | 1.0           | +0.05 / -0.00 | evaporator R th    |
| Post height = pocket depth | 3.0           | ±0.015        | lid bond / burst   |
| Rim + post-top flatness    | —             | 0.02          | hermeticity        |
| Groove depth               | 0.20          | ±0.03         | capillary return   |
| Bond face roughness        | Ra 0.4        | max           | diffusion bond     |
| Post position              | true position | Ø0.10 MMC     | lid land alignment |

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

CHAMBER BASE

Cu C1100 (OFHC) / 68 x 68 x 4.0 / 100 INTEGRAL POSTS

SCALE

2:1

UNITS

mm

THIRD ANGLE PROJECTION

DRAWING No.

VC-101

REV

A

SHEET

2 / 7

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

SEE NOTES

DATE

2026-09-09

**PLAN — WICK SIDE UP**  
SCALE 2:1

34

68

7

100 x Ø2.6

WICK 0.20 THK

60 WICK ZONE

68

**DETAIL E — POST LAND**  
SCALE 20:1

DIFFUSION BOND

1.0

0.20

Ø2.6

Ø2.0

**DETAIL F — FILL PORT**  
SCALE 10:1

TUBE OD3.0 / ID2.0 x 25 LG

BAG-8 BRAZE FILLET

0.30

**NOTES**

1. MATL: Cu C1100 OFHC annealed, 1.0 sheet.
2. Flatness 0.02 over 68 x 68 after sintering.
3. Wick clearance holes keep post lands bare for metal-to-metal bond.
4. Braze tube before bake-out, after diffusion bonding.
5. Handle with gloves. No hydrocarbons on bond faces.

1. MATL: Cu C1100 OFHC annealed, 1.0 sheet.
2. Flatness 0.02 over 68 x 68 after sintering.
3. Wick clearance holes keep post lands bare for metal-to-metal bond.
4. Braze tube before bake-out, after diffusion bonding.
5. Handle with gloves. No hydrocarbons on bond faces.

SCALE 20:1

Ø2.6

Ø2.0

D2.0 x 25 LG

10:1

BAg-8 BRAZE FILLET

0.30

Ø3.20

Ø1.20

**Pinch off, cold-weld, then laser seal the tip. Do NOT use a screw seal.**

|                                                                                        |  |                        |  |                                               |                       |                     |  |                    |  |
|----------------------------------------------------------------------------------------|--|------------------------|--|-----------------------------------------------|-----------------------|---------------------|--|--------------------|--|
| INDIA-VC THERMAL HARDWARE                                                              |  |                        |  |                                               | ISO 128 / ISO 2768-mK |                     |  |                    |  |
| TITLE<br><b>CHAMBER LID</b><br>Cu C1100 (OFHC) / 68 x 68 x 1.0 / WICK-LINED, FILL PORT |  |                        |  | SCALE<br><b>2:1</b><br>THIRD ANGLE PROJECTION |                       | UNITS<br><b>mm</b>  |  |                    |  |
| DRAWING No. <b>VC-102</b>                                                              |  |                        |  | REV <b>A</b>                                  |                       | SHEET <b>3 / 7</b>  |  |                    |  |
| DRAWN<br>AI-ASSISTED DRAFT                                                             |  | CHECKED<br>— PENDING — |  | MATERIAL<br>SEE BOM                           |                       | FINISH<br>SEE NOTES |  | DATE<br>2026-09-09 |  |

1

2

3

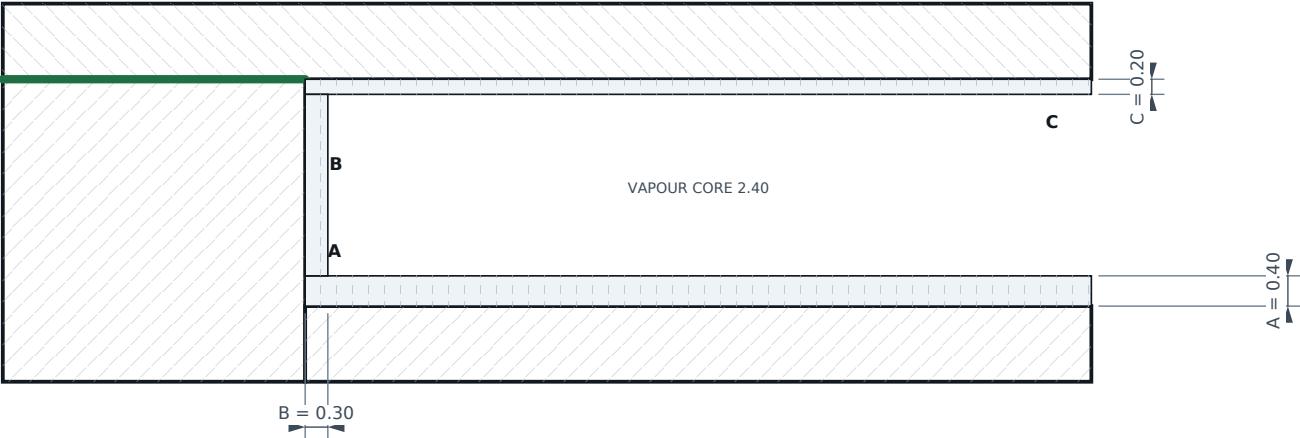
4

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6

WICK ZONE SECTION

SCALE 10:1 — ASSEMBLED



Zones A, B and C are sintered in ONE cycle and must be capillary-continuous:

condensate returns C → B → A. A break at the B/A junction causes evaporator dry-out.

| WICK ZONES |                             |      |                            |
|------------|-----------------------------|------|----------------------------|
| ZONE       | LOCATION                    | THK  | FUNCTION                   |
| A          | Pocket floor, around posts  | 0.40 | evaporator + liquid supply |
| B          | Pocket side walls (4 off)   | 0.30 | perimeter return path      |
| C          | Lid underside, posts masked | 0.20 | condensate collection      |

| SINTER PROCESS                     |                         |                                   |
|------------------------------------|-------------------------|-----------------------------------|
| PARAMETER                          | VALUE                   | TOL / NOTE                        |
| Cu powder, spherical, gas-atomised | 45 - 75 um              | sieved, -200/+325 mesh            |
| Effective pore radius              | 12.6 um                 | = 0.21 x d(particle)              |
| Porosity                           | 55 %                    | ± 5 %, quantitative metallography |
| Permeability (Blake-Kozeny)        | 1.97e-11 m2             | derived, not measured             |
| Capillary head, water @ 60 C       | 10.5 kPa                | vs 0.66 kPa gravity head          |
| Sintering temperature              | 950 C                   | ± 10 C                            |
| Dwell                              | 30 min                  | at temperature                    |
| Atmosphere                         | H2 (dew pt < -40 C)     | or vacuum < 1e-3 Pa               |
| Ramp / cool                        | 10 C/min / furnace cool | no quench                         |
| Tooling                            | graphite mandrel + mask | masks post tops, rim, fill port   |

WICK ACCEPTANCE

| ACCEPTANCE TEST       | REQUIREMENT       | METHOD                         |
|-----------------------|-------------------|--------------------------------|
| Porosity              | 55 % ± 5          | image analysis, 3 sections     |
| Wick thickness A      | 0.40 ± 0.05       | cross-section, 5 points        |
| Capillary rise, water | > 25 mm in 60 s   | witness coupon, vertical       |
| Groove continuity     | open, no blockage | back-light / CT, first article |

POWDER TRADE STUDY — WHY 45-75 um

| POWDER                    | r eff   | dP cap   | Q cap VERT | VERDICT           |
|---------------------------|---------|----------|------------|-------------------|
| 25 um                     | 5.2 um  | 25.2 kPa | 348 W      | clogs, low K      |
| 45 - 75 um                | 12.6 um | 10.5 kPa | 804 W      | SELECTED          |
| 100 um                    | 21.0 um | 6.3 kPa  | 1279 W     | coarse, ok        |
| 150 um                    | 31.5 um | 4.2 kPa  | 1802 W     | marginal tilt     |
| 400 um (original drawing) | 100 um  | 0.4 kPa  | 0 W        | FAILS — < gravity |

The original 200 um pore spec gives 380 Pa of capillary head against a 500 Pa gravity head

over 68 mm: the wick cannot lift liquid at all in any vertical orientation.

CAPILLARY PRESSURE BUDGET — WORST CASE, VERTICAL

| TERM                               | VALUE       | NOTE                      |
|------------------------------------|-------------|---------------------------|
| Capillary head available 2σ/r      | 10 508 Pa   | water @ 60 C, r = 12.6 um |
| Liquid loss through wick + grooves | – 12.6 Pa/W | at 804 W: 10 127 Pa       |
| Vapour loss through 2.40 core      | – 0.2 Pa/W  | negligible                |
| Gravity head, 68 mm adverse        | – 656 Pa    | vertical, evaporator up   |
| Margin at 113 W design point       | + 8 428 Pa  | 7.4 x                     |

The wick is sized by the vertical case. Horizontal operation carries 858 W and is never the limit.

PROCESS NOTES

- Mask post top faces and the 4.0 rim before sintering — sintered powder on a bond land destroys hermeticity.
- Mask the fill port bore. Powder in the pinch-off tube prevents a gas-tight cold weld.
- Grooves must remain open under the wick. Verify by back-light or CT on the first article.
- Sinter and diffusion bond may share one furnace cycle only if the bond fixture applies load after the sinter dwell; otherwise run separately.
- No organic binders. Any residue outgasses into the sealed chamber as non-condensable gas.

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

SINTERED WICK — PROCESS SPECIFICATION

IN-SITU SINTERED Cu POWDER / ALL WICK ZONES

SCALE

10:1

UNITS

mm

THIRD ANGLE PROJECTION

DRAWING No.

VC-103

REV

A

SHEET

4 / 7

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

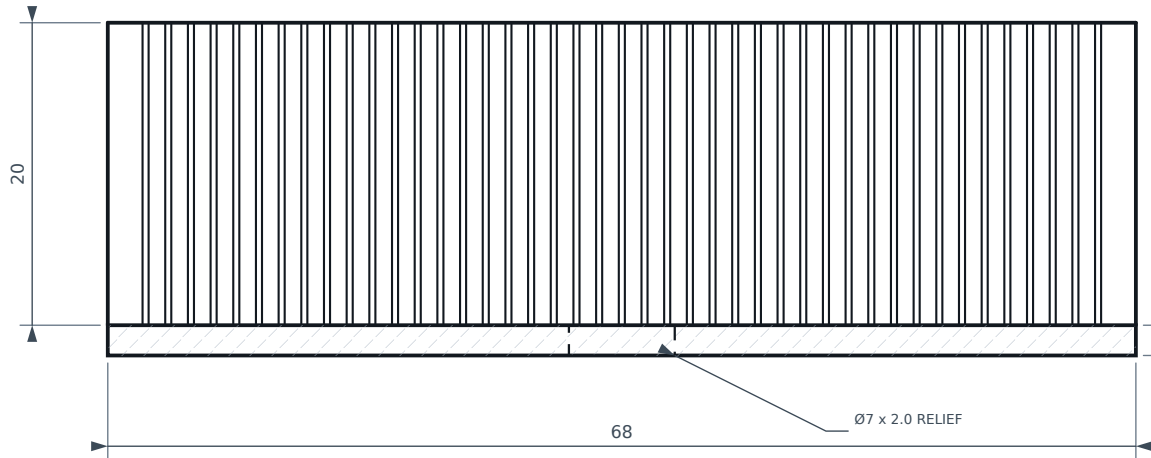
SEE NOTES

DATE

2026-09-09

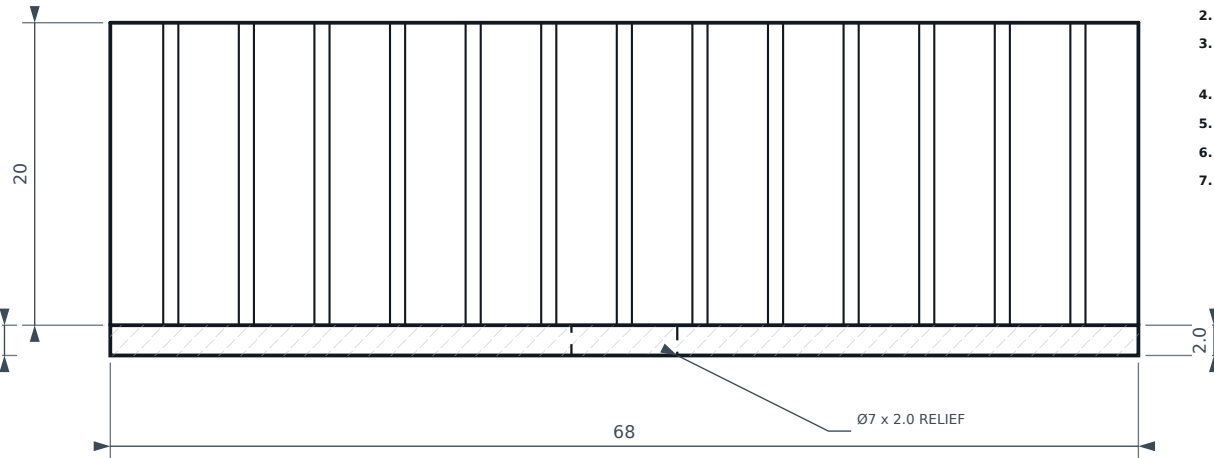
FORCED AIR  $\geq 1$  m/s

**43 FINS x 0.40 THK @ 1.5 PITCH**

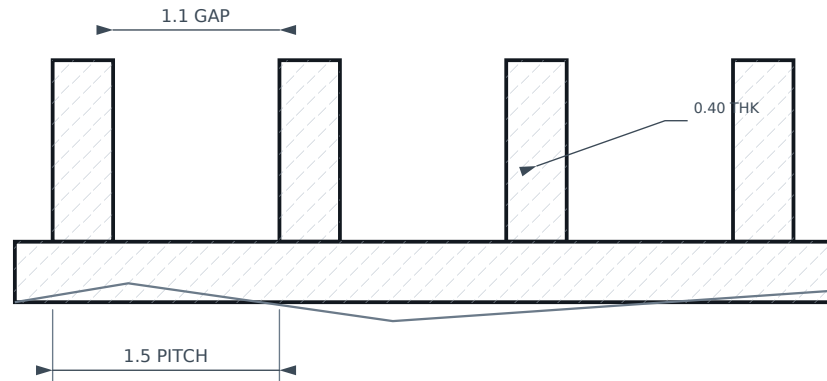


## NATURAL CONVECTION ONLY

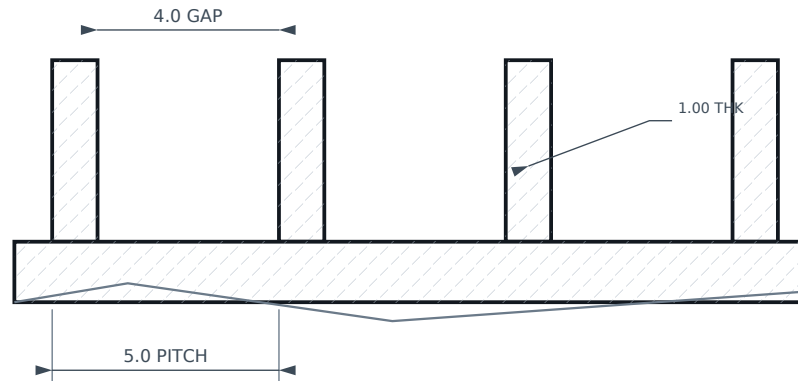
**13 FINS x 1.00 THK @ 5.0 PITCH**



SCALE 20:1



## SCALE 6:1



| AMBIENT | dT   | FIN BASE | CONVECTION | RADIATION | TOTAL  |
|---------|------|----------|------------|-----------|--------|
| 25 C    | 60 K | 82 C     | 11.9 W     | 4.6 W     | 16.5 W |
| 30 C    | 55 K | 82 C     | 10.5 W     | 4.3 W     | 14.8 W |
| 35 C    | 50 K | 82 C     | 9.1 W      | 4.0 W     | 13.1 W |
| 40 C    | 45 K | 83 C     | 7.7 W      | 3.7 W     | 11.4 W |
| 45 C    | 40 K | 83 C     | 6.4 W      | 3.4 W     | 9.8 W  |

| FIN FINISH                            | eps  | RAD   | TOTAL         | GAIN  |
|---------------------------------------|------|-------|---------------|-------|
| bare mill finish                      | 0.05 | 0.3 W | 12.4 W        | 1.00x |
| chromate conversion                   | 0.60 | 3.3 W | 15.2 W        | 1.23x |
| <b>clear anodise 5 um (SPECIFIED)</b> | 0.85 | 4.6 W | <b>16.5 W</b> | 1.33x |
| black anodise                         | 0.92 | 5.0 W | 16.9 W        | 1.36x |

Passive floor with channels fully blocked: 4.6 W from envelope radiation alone.

|                                        | PITCH      | FINS      | FINISH          | Q @ Tj 100 C | Q @ Tj 85 C   |
|----------------------------------------|------------|-----------|-----------------|--------------|---------------|
| <b>VC-201 - forced air 1 m/s</b>       | <b>1.5</b> | <b>43</b> | any             | <b>67 W</b>  | <b>54 W</b>   |
| <b>VC-201 - forced air 2 m/s</b>       | <b>1.5</b> | <b>43</b> | any             | <b>113 W</b> | <b>90 W</b>   |
| <b>VC-201 - forced air 4 m/s</b>       | <b>1.5</b> | <b>43</b> | any             | <b>163 W</b> | <b>130 W</b>  |
| VC-201 - STILL AIR (invalid)           | 1.5        | 43        | anodised        | 9 W          | 7 W           |
| <b>VC-202 - passive, fins VERTICAL</b> | <b>5.0</b> | <b>13</b> | <b>anodised</b> | <b>22 W</b>  | <b>16.5 W</b> |
| VC-202 - passive, fins horizontal      | 5.0        | 13        | anodised        | 13 W         | 9.5 W         |
| VC-202 - passive, bare aluminium       | 5.0        | 13        | bare            | 17 W         | 12.4 W        |

## ISO 128 / ISO 2768-mK

### THIRD ANGLE PROJECTION

SHEET 5 / 7

2026-09-09

## 1

2

32

## 3

- 4

Steps 7, 9 and 10 exist solely to prevent it. None of them is optional.

4

reading is not traceable once the chamber is cold and under vacuum.

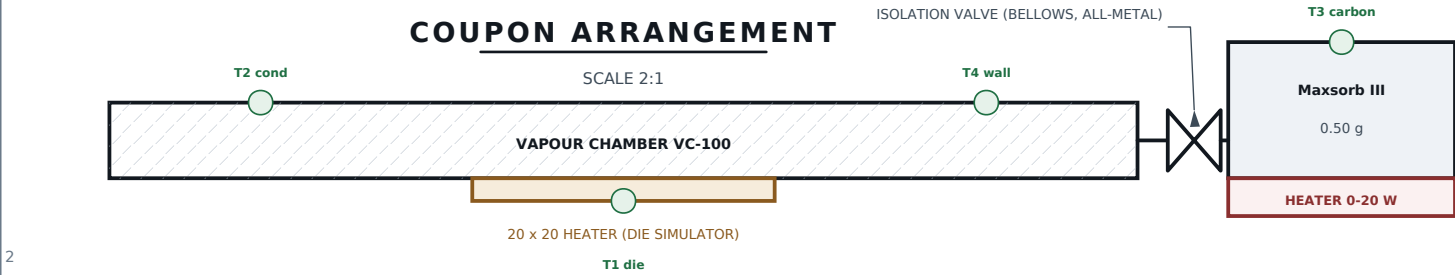
6



1

1

EXPERIMENTAL COUPON — NOT A PRODUCTION CONFIGURATION — SEE PHYSICS NOTE BELOW



Mesh: 10 um SUS 316L — NOT 50 um. Maxsorb III fines pass a 50 um mesh and foul the wick.

Carbon bake-out 250 C / 8 h / < 1e-3 Pa before installation, transferred under vacuum.

Reservoir 0.5 g Maxsorb III needs >= 1.8 cm3 at 0.28 g/cc bulk density — the original 1.50 cm3

housing could not physically hold the charge. This coupon uses 30 x 12 x 6 = 2.16 cm3.

| ADSORPTION EQUILIBRIUM — DUBININ-ASTAKHOV, METHANOL / MAXSORB III |       |        |          |        |           |          |
|-------------------------------------------------------------------|-------|--------|----------|--------|-----------|----------|
| T evap                                                            | T res | P/Psat | UPTAKE   | HELD   | WICK FILL | RESULT   |
| 60 C                                                              | 40 C  | 1.00   | 1.31 g/g | 0.66 g | 15 %      | DRY      |
| 60 C                                                              | 60 C  | 1.00   | 1.31 g/g | 0.66 g | 15 %      | DRY      |
| 60 C                                                              | 100 C | 0.24   | 1.06 g/g | 0.53 g | 35 %      | DRY      |
| 60 C                                                              | 127 C | 0.11   | 0.77 g/g | 0.39 g | 58 %      | marginal |
| 60 C                                                              | 169 C | 0.03   | 0.42 g/g | 0.21 g | 90 %      | wet      |

4

4

PHYSICS NOTE — WHY THE PASSIVE VERSION CANNOT WORK

- In a sealed pure-substance chamber the pressure is set by the evaporator:  $P = P_{\text{sat}}(T_{\text{evap}})$ . The carbon sees the relative pressure  $P / P_{\text{sat}}(T_{\text{res}})$ .
- A reservoir on the condenser side always satisfies  $T_{\text{res}} \leq T_{\text{evap}}$ , so  $P/P_{\text{sat}} \geq 1$  and the carbon saturates: it takes 0.66 g of a 0.75 g charge and the wick runs at 15 % fill. That is permanent dry-out, not pressure control.
- Keeping the wick wet needs the reservoir ~110 K ABOVE the chip. A working vapour chamber is isothermal, so no passive layout can deliver that. It requires the heater and valve drawn above.
- Overcharging to 1.90 mL keeps the wick wet, but then liquid always remains, the pressure stays at  $P_{\text{sat}}$ , and the carbon buys nothing thermally. Its only genuine residual function is as a non-condensable gas getter — for which tens of mg suffices, not 0.5 g.

6

6

| COUPON vs PRODUCTION VC-100 |                          |          |
|-----------------------------|--------------------------|----------|
|                             | WITH CARBON              | WITHOUT  |
| Wick fill at equilibrium    | 15 %                     | 125 %    |
| Q at 2 m/s forced air       | 0 W (dry)                | 113 W    |
| Transient buffer            | 722 J (heater needed)    | 0 J      |
| Parts count / failure modes | + valve, heater, control | baseline |

| INSTRUMENTATION  |                              |                            |
|------------------|------------------------------|----------------------------|
| MEASUREMENT      | INSTRUMENT                   | PURPOSE                    |
| T1..T4           | type-T TC, $\pm 0.2$ K       | gradient map               |
| Chamber pressure | capacitance manometer        | verify $P_{\text{sat}}(T)$ |
| Reservoir mass   | in-situ load cell $\pm 1$ mg | uptake vs time             |
| Heater power     | 4-wire, $\pm 0.5$ %          | energy balance             |
| Valve state      | open / closed                | isolate the carbon         |

| TEST MATRIX |                                              |                                        |
|-------------|----------------------------------------------|----------------------------------------|
| #           | TEST                                         | MEASURES                               |
| 1           | Valve CLOSED, sweep 0-150 W                  | baseline R th, no carbon               |
| 2           | Valve OPEN, reservoir unheated               | dry-out rate + time constant           |
| 3           | Valve OPEN, reservoir held 110 K above die   | whether the wick recovers              |
| 4           | Valve OPEN, cyclic 10 W bursts, heater regen | usable buffer energy (722 J predicted) |
| 5           | 1000 h hold, valve OPEN, unheated            | NCG getter benefit vs dry-out cost     |

5

5

COUPON NOTES

- Fluid is METHANOL for this coupon (activated carbon is hydrophobic; water uptake is poor). Accept the 7.3x lower merit number — this coupon measures adsorption, not peak cooling.
- Methanol charge 1.90 mL so the wick stays wet with the carbon saturated. Do not run at 1.00 mL.
- The valve is what makes this an experiment rather than a defect. A production unit has no valve, so it has no way to stop the carbon draining the wick.

5

5

IF TEST 4 SUCCEEDS — PATH TO A PRODUCT

- Move the reservoir onto the die's own heat path, upstream of the chamber, so it is genuinely hotter than the evaporator without a valve.
- Replace the bellows valve with a thermally actuated one-way element, or accept a control loop driving the regeneration heater.
- Budget 722 J against ~5 g of added mass and a new hermetic joint. Compare against simply adding 15 K of copper (+50 g).

| INDIA-VC THERMAL HARDWARE                        |             |                        |           | ISO 128 / ISO 2768-mK |       |
|--------------------------------------------------|-------------|------------------------|-----------|-----------------------|-------|
| TITLE                                            |             | SCALE                  |           | UNITS                 |       |
| MAXSORB III RESERVOIR — TEST COUPON              |             | 2:1                    |           | mm                    |       |
| EXPERIMENTAL / NOT FOR PRODUCTION / INSTRUMENTED |             | THIRD ANGLE PROJECTION |           |                       |       |
| DRAWING No.                                      |             | VC-400                 |           | REV                   | A     |
|                                                  |             |                        |           | SHEET                 | 7 / 7 |
| DRAWN                                            | CHECKED     | MATERIAL               | FINISH    | DATE                  |       |
| AI-ASSISTED DRAFT                                | — PENDING — | SEE BOM                | SEE NOTES | 2026-09-09            |       |

6

6